MATH 110 — Week 6

Notes Chapter 11; Homework 6

§1 Invariant subspaces and eigenvectors

The question. Which subspaces remain inside themselves when an operator acts? For $T(x, y) = (x, 2y)$, each coordinate axis is preserved.

Definition (Notes Definition 30; LADR 5.1–5.2)**.** An operator $T \in L(V)$ is a linear map $T : V \to V$. A subspace $U \subseteq V$ is invariant under an operator $T \in L(V)$ if $T(U) \subseteq U$. Its restriction $T|_U : U \to U$ is then an operator. By Week 5 Proposition 42 it also induces an operator $T/U : V/U \to V/U$, $(T/U)(v + U) = Tv + U$.

Proposition (Testing invariance on generators)
If $U = \operatorname{span}(u_1, \ldots, u_k)$, then U is invariant under T iff $Tu_j \in U$ for every j.

Proof. 1. Invariance implies the stated containment because each u_j belongs to U.
2. Conversely, $T(\sum_j a_j u_j) = \sum_j a_j T u_j \in U$ by linearity and closure of U, proving invariance.

□

Example (Notes Example 48; LADR 5.4) 1. The spaces $\{0\}$ and V are invariant: $T0 = 0$, and every output belongs to V.
2. If $v \in \ker T$, then $Tv = 0 \in \ker T$, so the kernel is invariant.
3. If $w = Tv \in \operatorname{range} T$, then $Tw = T(Tv) \in \operatorname{range} T$, so the range is invariant.

Example (LADR 5.3, 5.6) 1. Differentiation on $\mathcal{P}(\mathbb{R})$ preserves $\mathcal{P}_4(\mathbb{R})$ because differentiating a degree-at-most 4 polynomial gives degree at most 3.
2. For $T(x, y, z) = (7x + 3z, 3x + 6y + 9z, -6y)$, direct substitution gives $T(3, 1, -1) = (18, 6, -6) = 6(3, 1, -1)$. Thus the line spanned by $(3, 1, -1)$ is invariant.

Definition (Notes Definitions 31–32; LADR 5.5, 5.8, 5.52)**.** A nonzero $v \in V$ is an eigenvector if $Tv = \lambda v$ for some scalar $\lambda \in F$; that scalar is its eigenvalue. Equivalently the line Fv is invariant. The scalar is unique because $(\lambda - \mu)v = 0$ and $v \neq 0$ force $\lambda = \mu$. For any scalar λ, the eigenspace is $E(\lambda, T) = \ker(T - \lambda I)$. It contains zero, and its nonzero elements are precisely the eigenvectors with eigenvalue λ.

Proposition (Notes Definition 32; LADR 5.7)
$E(\lambda, T)$ is an invariant subspace. For finite-dimensional V, the following are equivalent: λ is an eigenvalue; $T - \lambda I$ is not injective; $T - \lambda I$ is not surjective; $T - \lambda I$ is not invertible.

Proof. 1. A kernel is a subspace by Week 3. If $v \in E(\lambda, T)$, then $Tv = \lambda v$ belongs to the same subspace, proving invariance.

2. The equation $Tv = \lambda v$ for nonzero v is equivalent to $(T - \lambda I)v = 0$ for nonzero v, precisely failure of injectivity.

3. Rank–nullity for $T - \lambda I$ says $\dim V = \dim \ker(T - \lambda I) + \dim \operatorname{range}(T - \lambda I)$. Zero kernel is therefore equivalent to full range, and a linear bijection has a linear inverse (Week 4 Proposition 31). This proves the remaining equivalences.

□

Example (Notes Example 49; quarter-turn: LADR 5.9)

Every nonzero proper subspace of $\mathbb{R}^2$ is a line by Week 2; invariant lines therefore correspond to real eigenvectors.

1. For $A = \operatorname{diag}(1, 2)$, $A(x, y) = \lambda(x, y)$ gives $(1 - \lambda)x = 0$ and $(2 - \lambda)y = 0$. A nonzero solution has either $\lambda = 1, y = 0$ or $\lambda = 2, x = 0$. Thus the invariant subspaces are $\{0\}, \mathbb{R}e_1, \mathbb{R}e_2, \mathbb{R}^2$.
2. For $R = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, $R^2 = -I$. An eigenvector would satisfy $\lambda^2 v = -v$, hence $\lambda^2 = -1$, impossible over $\mathbb{R}$. Only $\{0\}, \mathbb{R}^2$ are invariant.
3. For $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $N^2 = 0$, so an eigenvalue satisfies $\lambda^2 = 0$ and hence $\lambda = 0$. Its kernel and range are both $\mathbb{R}e_1$: $N(x, y) = (y, 0)$. Thus the invariant subspaces are $\{0\}, \mathbb{R}e_1, \mathbb{R}^2$; $\mathbb{R}e_2$ is not invariant because $Ne_2 = e_1$.

$$e_2 \xrightarrow{N} e_1 \xrightarrow{N} 0$$

The chain $\{0\} \subset \mathbb{R}e_1 \subset \mathbb{R}^2$ consists of invariant subspaces.

Definition (Notes Chapter 11: triangular matrix convention)**.** A square matrix is upper triangular when every entry below its main diagonal is zero. For an ordered basis $v_1, \ldots, v_n$, put $U_j = \operatorname{span}(v_1, \ldots, v_j)$.

Proposition (Notes Chapter 11, invariant initial-span observation)

The matrix of T in the ordered basis $v_1, \ldots, v_n$ is upper triangular if and only if every U_j is invariant under T.

Proof. 1. If every U_j is invariant, then $Tv_j \in U_j$. Its coordinates in positions $i > j$ vanish, so the entries below the diagonal in column j are zero.

2. Conversely, an upper-triangular matrix has $Tv_k \in U_k \subseteq U_j$ for $k \leq j$. Applying T to any linear combination of $v_1, \ldots, v_j$ therefore stays in U_j.

□

A probability column has nonnegative real entries summing to 1. A column-stochastic matrix has nonnegative entries with each column summing to 1. It preserves probability columns: $(Pp)_i = \sum_j P_{ij} p_j \geq 0$ and $\sum_i (Pp)_i = \sum_j (\sum_i P_{ij}) p_j = \sum_j p_j = 1$ (Week 3).

Definition (Notes Chapter 11: stationary distribution)**.** A probability vector π is stationary for a column-stochastic matrix P when $P\pi = \pi$. It is an eigenvector with eigenvalue 1, because its entries sum to 1 and therefore it is nonzero. Induction gives $P^k \pi = \pi$ for all $k \geq 0$.

Example (Notes Chapter 11: the three-state distribution)

For $P = \begin{pmatrix} 1/2 & 1/4 & 0 \\ 1/2 & 1/2 & 1/2 \\ 0 & 1/4 & 1/2 \end{pmatrix}$, let $\pi = (1/4, 1/2, 1/4)^{\mathsf{T}}$.

1. The product is $P\pi = (1/8 + 1/8,\ 1/8 + 1/4 + 1/8,\ 1/8 + 1/8)^{\mathsf{T}} = (1/4, 1/2, 1/4)^{\mathsf{T}} = \pi$. Its nonnegative entries sum to 1, so it is a stationary distribution and an eigenvector with eigenvalue 1.
2. Existence, uniqueness, convergence from other starts, and the convergence rate are separate questions. For example I_2 fixes every distribution, giving nonuniqueness.
3. The swap matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ fixes the unique probability vector $(1/2, 1/2)^{\mathsf{T}}$, since its fixed vectors have equal coordinates. Yet it alternates e_1, e_2 forever, showing failure of convergence.

Shortcuts

- Test invariance on a subspace basis: linearity extends containment from those images to every vector.
- An invariant line is an eigenvector line; in dimension 2 this classifies all proper nonzero invariant subspaces.

Traps

- An eigenspace contains zero, but zero is never an eigenvector by definition.
- An eigenvalue depends on the field: the quarter-turn has no real eigenvalue, while over $\mathbb{C}$, $R(1,-i) = i(1,-i)$ gives an eigenvector.
- Invariant does not mean fixed pointwise: $\mathbb{R}e_2$ is invariant under $\operatorname{diag}(1,2)$ but e_2 is sent to $2e_2$.

§2 Complex roots and the fundamental theorem of algebra

The question. For $z = a + bi$, write $|z| = \sqrt{a^2+b^2}$. Why can a complex direction reveal a root that real inputs miss? For $q(z) = 1 + z^2$, $|q(t)| = 1 + t^2$ on the real axis, but $|q(it)| = 1 - t^2 < 1$ when $0 < t < 1$.

Definition (Notes Chapter 11: modulus and polar form)**.** For $z = a + bi$, let $\overline{z} = a - bi$ and $|z| = \sqrt{a^2+b^2} = \sqrt{z\overline{z}}$. Write $\operatorname{Re} z = a$ and $\operatorname{Im} z = b$. A nonzero complex number has polar form $z = r(\cos\theta + i\sin\theta)$, with $r = |z| > 0$. The angle θ is called an argument of z; the real pair $(a/r, b/r)$ lies on the unit circle. We cite the elementary parametrization of that circle, the trigonometric addition identities, and the existence of positive real kth roots (from continuity of $t \mapsto t^k$).

Proposition (Complex arithmetic tools for Notes Proposition 52)

$|zw| = |z||w|$ and $|z + w| \le |z| + |w|$. Every nonzero complex number has a kth root for each positive integer k.

Proof. 1. For $z = a+bi, w = c+di$, multiplication gives $\overline{zw} = (ac-bd)-i(ad+bc) = \overline{z}\,\overline{w}$, and $|\overline{w}| = \sqrt{c^2+(-d)^2} = |w|$. Hence $|zw|^2 = zw\overline{zw} = |z|^2|w|^2$; both sides are nonnegative, giving multiplicativity. Also $\operatorname{Re} u \le |u|$ because $|u|^2 = (\operatorname{Re} u)^2 + (\operatorname{Im} u)^2$.

2. $|z+w|^2 = |z|^2 + 2\operatorname{Re}(z\overline{w}) + |w|^2 \le |z|^2 + 2|z||w| + |w|^2$. Taking square roots proves the triangle inequality; iteration gives it for finite sums. Also $|z| \le |z+w| + |w|$ gives the reverse bound $|z+w| \ge |z| - |w|$.

3. By induction using the addition identities, $(\cos\phi + i\sin\phi)^k = \cos(k\phi) + i\sin(k\phi)$. Thus if $z = r(\cos\theta + i\sin\theta)$, then $r^{1/k}(\cos(\theta/k) + i\sin(\theta/k))$ has kth power z.

□

Compute a complex square root

1. The number i has modulus 1 and argument $\pi/2$, so one square root is $\cos(\pi/4) + i\sin(\pi/4) = (1+i)/\sqrt{2}$.
2. Directly, $((1+i)/\sqrt{2})^2 = (1 + 2i + i^2)/2 = i$, verifying the result.

Analytic tools used in the next proof (cited). A continuous real function on a closed bounded disk in $\mathbb{R}^2$ attains its minimum (the extreme-value theorem). Polynomials and modulus are continuous by elementary real analysis. These are the analytic inputs; the decrease from a nonzero minimum is proved explicitly below.

Proposition (Notes Proposition 52; LADR 4.12: fundamental theorem of algebra)

Every nonconstant polynomial with complex coefficients has a complex root.

Where the proof comes from

Want. Force the smallest polynomial modulus to be zero.

Know. The leading term dominates for large modulus, and a continuous function attains a minimum on a closed disk.

Key step. At a hypothetical nonzero minimum, normalize to $q(z) = 1 + a_k z^k + \cdots$ with $a_k \neq 0$. Choose $a_k \beta^k = -1$, giving $q(t\beta) = 1 - t^k +$ remainder. The proof bounds that remainder by Ct^{k+1}; choosing $Ct < 1$ makes it smaller than the decrease t^k.

Proof. 1. Write $p(z) = \sum_{j=0}^{m} c_j z^j$, $c_m \neq 0$, $m \geq 1$. For $|z| \geq 1$, put $C_0 = \sum_{j<m} |c_j|$; the triangle inequality gives $|p(z)| \geq |c_m||z|^m - C_0|z|^{m-1}$. For sufficiently large $|z|$ this tends to infinity.

2. Choose a closed disk outside which $|p(z)| > |p(0)|$. On the disk $|p|$ attains a minimum at ζ by the cited theorem; since 0 is in the disk, this is a global minimum.

3. Suppose $p(\zeta) \neq 0$. Define $q(z) = p(\zeta + z)/p(\zeta) = 1 + a_k z^k + \cdots + a_m z^m$, with $a_k \neq 0$ and $k \geq 1$. The minimum condition gives $|q(z)| \geq 1$ for every z.

4. Choose β with $\beta^k = -1/a_k$, using the proved root tool. For $0 < t < 1$, $q(t\beta) = 1 - t^k + \sum_{j>k} a_j \beta^j t^j$; the remainder has modulus at most Ct^{k+1}, where $C = \sum_{j>k} |a_j \beta^j|$.

5. Choose $0 < t < 1$ also satisfying $Ct < 1$. Then $1 - t^k > 0$, and $|q(t\beta)| \leq 1 - t^k + Ct^{k+1} = 1 - t^k(1 - Ct) < 1$, a contradiction. Thus $p(\zeta) = 0$.

□

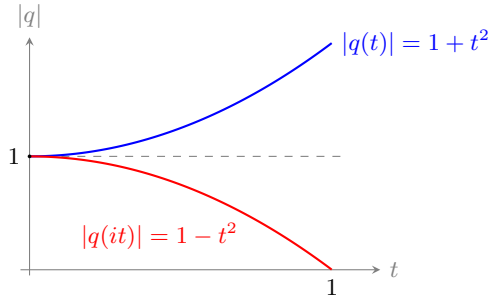
Proposition (Notes Chapter 11; LADR 4.13: complex factorization)

If p has degree m over $\mathbb{C}$, then $p(z) = c \prod_{j=1}^{m} (z - \lambda_j)$, with $c \neq 0$ and roots repeated as needed. This property is called algebraic closure of $\mathbb{C}$.

Proof. 1. If $p(\lambda) = 0$, the identity $z^r - \lambda^r = (z - \lambda) \sum_{j=0}^{r-1} z^{r-1-j} \lambda^j$ applied to $p(z) - p(\lambda)$ gives $p = (z - \lambda)q$ of degree $m - 1$.

2. Apply Proposition 52 to each nonconstant remaining factor. After m steps only a nonzero constant c remains, proving the formula.

□



For $q(z) = 1 + z^2$ and $0 < t < 1$, real inputs increase the modulus above 1, while imaginary inputs decrease it below 1. Thus 0 cannot be a minimum over the complex plane.

Shortcuts

- To find a kth root of a nonzero complex number, use polar form and divide an argument by k.
- The first nonzero term after a constant controls the local decrease in the FTA proof; the higher terms are bounded by one extra factor of t.

Traps

- A nonconstant real polynomial need not have a real root: $x^2 + 1 > 0$ for all real x.
- Minimizing only over real directions is insufficient: for $1 + z^2$, real motion increases the modulus at 0, but imaginary motion decreases it.
- The theorem asserts existence, not a formula in radicals. Such formulas fail for general degree 5 polynomials (a cited algebraic result), without contradicting root existence.

§3 Polynomials of operators and existence of eigenvectors

The question. Can a polynomial identity constrain eigenvalues? If $T^2 = I$ and $Tv = \lambda v$ with $v \neq 0$, then $\lambda^2 v = v$, so $\lambda^2 = 1$.

Definition (LADR 5.13–5.14, 5.16)**.** For an operator T, $T^0 = I$ and T^k is the repeated composition for $k \geq 1$. If T is invertible, define $T^{-k} = (T^{-1})^k$. For $p(z) = \sum_{j=0}^{m} a_j z^j$, define $p(T) = \sum_{j=0}^{m} a_j T^j$. Polynomial multiplication is ordinary multiplication and collection of equal powers: the coefficient of z^r in pq is $\sum_{j+k=r} a_j b_k$.

Proposition (LADR 5.13, 5.17)

Powers satisfy $T^j T^k = T^{j+k}$ and $(T^j)^k = T^{jk}$ for nonnegative integers, and for all integers when T is invertible. Polynomial evaluation satisfies $(p + q)(T) = p(T) + q(T)$, $(ap)(T) = ap(T)$, and $(pq)(T) = p(T)q(T) = q(T)p(T)$.

Where the proof comes from

Want. Use ordinary polynomial algebra for expressions in one operator.

Know. Composition is associative and distributive, and $TT^{-1} = T^{-1}T = I$ when defined.

Key step. Expand the finite sums and collect powers of T.

Proof. 1. For nonnegative powers, concatenating the repeated compositions gives $T^j T^k = T^{j+k}$. For negative powers, cancel adjacent pairs TT^{-1} or $T^{-1}T$ until only the required power remains. Grouping k copies of T^j gives $(T^j)^k = T^{jk}$ for nonnegative exponents; the same cancellation proves the integer case when T is invertible.

2. The addition and scalar identities follow coefficientwise. For products, $p(T)q(T) = \sum_{j,k} a_j b_k T^{j+k} = \sum_r (\sum_{j+k=r} a_j b_k) T^r = (pq)(T)$.

3. Polynomials commute under multiplication over the field, so $(pq)(T) = (qp)(T) = q(T)p(T)$.

□

Example (LADR 5.15: compute a polynomial of differentiation)

Let $Dq = q'$ and $p(z) = 7 - 3z + 5z^2$.

1. By definition, $p(D)q = 7q - 3q' + 5q''$.
2. For $q(x) = x^2$, the power rule gives $p(D)q = 7x^2 - 6x + 10$.

Proposition (LADR 5.18; polynomial action on eigenvectors)

For $S, T \in L(V)$ with $ST = TS$, $\ker S$ and $\operatorname{range} S$ are invariant under T. In particular, $\ker p(T)$ and $\operatorname{range} p(T)$ are invariant under T. If $Tv = \lambda v$, then $p(T)v = p(\lambda)v$. In particular, if $p(T) = 0$, every eigenvalue is a root of p.

Proof. 1. If $ST = TS$ and $Sv = 0$, then $S(Tv) = T(Sv) = 0$. Thus T preserves $\ker S$.
2. If $w = Sv$, then $Tw = TSv = STv \in \operatorname{range} S$. Take $S = p(T)$, which commutes with T by the polynomial product formula, to obtain the stated polynomial cases.
3. Induction gives $T^j v = \lambda^j v$ for $j \geq 0$. Hence $p(T)v = \sum a_j \lambda^j v = p(\lambda)v$.
4. If also $p(T) = 0$ and $v \neq 0$, then $p(\lambda)v = 0$ forces $p(\lambda) = 0$.

□

Proposition (Notes Proposition 53; LADR 5.19)

Every operator on a nonzero finite-dimensional complex vector space has an eigenvector.

Where the proof comes from

Want. Find a nonzero vector satisfying a first-degree operator equation.

Know. A list longer than the dimension is dependent, and every nonconstant complex polynomial has a root.

Key step. A relation $p(T)v = 0$ is not yet an eigenvector equation. After factoring $p(z) = (z - \lambda)q(z)$, the candidate is $w = q(T)v$; it must be nonzero. Choosing the first dependence makes every lower-degree $q(T)v$ nonzero, while the factorization gives $(T - \lambda I)w = 0$.

Proof. 1. Let $n = \dim V$ and choose $v \neq 0$. The list $v, Tv, \ldots, T^n v$ is dependent, so choose a nonzero polynomial p of smallest degree with $p(T)v = 0$. Its degree is at least 1 because a nonzero scalar multiple of v is nonzero.
2. By Proposition 52, p has a root λ, so $p(z) = (z - \lambda)q(z)$ with nonzero q of smaller degree.
3. Put $w = q(T)v$. Minimality of the degree of p gives $w \neq 0$.
4. By the polynomial product identity, $(T - \lambda I)w = p(T)v = 0$. Hence $Tw = \lambda w$, proving that w is an eigenvector.

□

Example (Notes Chapter 11; LADR 5.20: necessary hypotheses) 1. Over $\mathbb{R}$, the quarter-turn matrix has no eigenvector by Example 49, despite its finite dimension.
2. On $\mathcal{P}(\mathbb{C})$, $Tp = xp$ has no eigenvector: if $xp = \lambda p$ for nonzero p, then $(x - \lambda)p = 0$, impossible because its leading coefficient is nonzero and its degree is one larger.
3. The zero-dimensional space has no nonzero vector, so it has no eigenvector. These examples explain all three hypotheses in Proposition 53.

Proposition (Notes Proposition 54; LADR 5.11)

Eigenvectors for pairwise distinct eigenvalues are linearly independent. Consequently, the sum of the eigenspaces for distinct eigenvalues is direct.

Where the proof comes from

Want. Rule out a relation among vectors scaled by different factors.

Know. The first vector dependent on its predecessors has independent predecessors (Week 2 linear-dependence lemma).

Key step. A dependence does not yet isolate any coefficient. Applying T scales its terms by different λ_j; subtracting λ_k times the original relation removes v_k. The remaining independent predecessors have coefficients $a_j(\lambda_j - \lambda_k)$, whose nonzero eigenvalue differences force $a_j = 0$.

Proof. 1. Suppose the eigenvectors $v_1, \dots, v_m$ are dependent. Let k be the first index with $v_k = \sum_{j<k} a_j v_j$; the preceding vectors are independent. Since each eigenvector is nonzero, $k \geq 2$.
2. Apply T to obtain $\lambda_k v_k = \sum_{j<k} a_j \lambda_j v_j$. Subtract λ_k times the original relation to get $\sum_{j<k} a_j(\lambda_j - \lambda_k)v_j = 0$.
3. Independence forces $a_j(\lambda_j - \lambda_k) = 0$. All eigenvalue differences are nonzero, so every $a_j = 0$, giving $v_k = 0$, a contradiction.
4. In a zero sum of vectors from distinct eigenspaces, discard zero summands. Every remaining summand is an eigenvector with a distinct eigenvalue; their independence forces that list to be empty. This is the zero-relation criterion for a direct sum.

□

Example (Notes Example 50)

A smooth function on $\mathbb{R}$ has derivatives of every order. On their real vector space, let $Tf = f''$.

1. Elementary calculus gives $T(\sin(jx)) = -j^2 \sin(jx)$ for $j \geq 1$. Each function is nonzero, for example at $x = \pi/(2j)$.
2. The scalars $-1^2, \dots, -N^2$ are distinct. Proposition 54 therefore proves $\sin x, \sin 2x, \dots, \sin Nx$ independent for every positive integer N.

Proposition (Notes Proposition 55; LADR 5.12)

An operator on a finite-dimensional space has at most $\dim V$ distinct eigenvalues.

Proof. 1. Choose one nonzero eigenvector for each of any m distinct eigenvalues. Proposition 54 makes the list independent.
2. Week 2 gives $m \leq \dim V$. Thus more than $\dim V$ distinct eigenvalues are impossible.

□

Example (Notes Chapter 11: infinite-dimensional cardinality)

For the shift $S(a_0, a_1, \dots) = (a_1, a_2, \dots)$ on real sequences, each $\lambda \in \mathbb{R}$ gives eigenvector $v_\lambda = (1, \lambda, \lambda^2, \dots)$; at $\lambda = 0$ this means $(1, 0, 0, \dots)$.

1. Coordinate comparison gives $Sv_\lambda = \lambda v_\lambda$, and the first coordinate makes $v_\lambda \neq 0$.
2. Every finite list with distinct parameters is independent by Proposition 54. Thus the whole indexed family is independent, since a dependence is a finite relation.
3. The basis-extension/choice principle cited in Week 2 extends this independent family to a basis. Hence the dimension as a cardinal is at least the cardinality of $\mathbb{R}$, a stronger conclusion than merely infinite dimension.

Shortcuts

- An identity such as $T^2 = I$ restricts eigenvalues by replacing T with λ on an eigenvector.
- Distinct eigenvalues prove independence without solving a dependence system.

Traps

- $p(T)v = 0$ does not generally make v an eigenvector: $T = \operatorname{diag}(1, 2)$, $p(z) = (z-1)(z-2)$ and $v = (1, 1)$ give $p(T)v = 0$, but $Tv = (1, 2)$ is not a scalar multiple of v.
- A root of a relation $p(T) = 0$ need not occur as an eigenvalue: $T = I$ satisfies $(T - I)(T - 2I) = 0$, but 2 is not an eigenvalue over $\mathbb{R}$.
- Polynomial operators in one T commute, but arbitrary operators need not: Week3 gave $DX - XD = I$.

§4 Computing eigenvalues and eigenspaces

The question. How do we turn an operator equation into coordinate equations? For $T(x,y) = (x, 2y)$ and $\lambda = 1$, the equation $Tv = v$ gives $y = 0$. In general the coordinate equation is $(M(T) - \lambda I)[v] = 0$.

Proposition (Notes Proposition 56)

Let $T \in L(V)$ with V finite-dimensional, and fix an ordered basis of V. Then $Tv = \lambda v$ with $v \neq 0$ iff $M(T)[v] = \lambda[v]$ with $[v] \neq 0$. Thus the operator and its matrix have the same eigenvalues and corresponding eigenspaces under the coordinate isomorphism.

Proof. 1. Unique basis coefficients make $v \mapsto [v]$ a linear bijection: an output coordinate column specifies exactly one vector. Thus $Tv = \lambda v$ iff $[Tv] = [\lambda v]$.
2. The two sides equal $M(T)[v]$ and $\lambda[v]$, respectively. Also $v \neq 0$ iff $[v] \neq 0$, completing the equivalence.

□

Definition (Notes Chapter 11: two-by-two characteristic polynomial)**.** For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, define its characteristic polynomial by $\chi_A(t) = \det(tI - A)$, using only the two-by-two determinant from Week 4. Expanding gives $\chi_A(t) = (t-a)(t-d) - bc = t^2 - (a+d)t + (ad - bc)$.

Proposition (Notes Chapter 11; HW6 P4)

For a two-by-two matrix, $\lambda \in F$ is an eigenvalue iff $\chi_A(\lambda) = 0$.

Proof. 1. $Av = \lambda v$ for nonzero v iff $(\lambda I - A)v = 0$ for nonzero v, so $\lambda I - A$ is not injective and hence not invertible.
2. The two-by-two inverse criterion says this occurs exactly when $\det(\lambda I - A) = \chi_A(\lambda) = 0$.

□

Example (Notes Example 51) 1. For $R = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, $\chi_R(t) = t^2 + 1$. There are no real roots. Over $\mathbb{C}$ the roots are $i, -i$, and solving $Rv = \lambda v$ gives $E(i, R) = \mathbb{C}(1, -i)$, $E(-i, R) = \mathbb{C}(1, i)$.
2. For $F = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, $\chi_F(t) = t^2 - t - 1$. Completing the square gives roots $\lambda_\pm = (1 \pm \sqrt{5})/2$.
3. The equation $F(x,y) = \lambda(x,y)$ gives $y = \lambda x$, with the second equation following from $\lambda^2 = \lambda + 1$. Thus $E(\lambda_\pm, F) = \mathbb{R}(1, \lambda_\pm)$.

Example (Notes Example 52: transpose operator)

On $\mathbb{C}^{2,2}$, let $T(A) = A^\mathsf{T}$. Write E_{ij} for the matrix with a single 1 at entry (i,j) and zeros elsewhere.

1. Since $T^2 = I$, any eigenvalue satisfies $\lambda^2 = 1$, so it is 1 or -1.
2. The condition $A^\mathsf{T} = A$ says $A = \left(\begin{smallmatrix} a & b \\ b & d \end{smallmatrix}\right)$, the symmetric matrices. Hence $E(1, T)$ has basis $E_{11}, E_{12} + E_{21}, E_{22}$; comparing entries proves independence.
3. The condition $A^\mathsf{T} = -A$ forces the diagonal entries zero and the off-diagonal entries opposite. These skew-symmetric matrices form $E(-1, T) = \mathbb{C}(E_{12} - E_{21})$. Both eigenspaces are nonzero, so both possible eigenvalues occur.

Solve a repeated-eigenvalue example

1. For $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, $\chi_A(t) = (t-1)^2$, so the only eigenvalue is 1.
2. Solving $(A - I)(x,y) = (y, 0) = 0$ gives $E(1, A) = \mathbb{R}e_1$.
3. An eigenvector basis would need two independent vectors, but all eigenvectors lie in this one-dimensional space. Therefore no basis of eigenvectors exists.

Shortcuts

- After finding each scalar eigenvalue, solve the homogeneous kernel equations; roots alone do not determine the eigenvectors.
- Use a relation $p(T) = 0$ or a visible invariant subspace before expanding a characteristic polynomial.

Traps

- A repeated root does not guarantee enough eigenvectors: the matrix $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ above has only one independent eigenvector.
- Over $\mathbb{R}$, a quadratic with no real roots gives no eigenvectors for a two-by-two operator; using complex roots silently changes the field.

§5 PageRank and stochastic eigenvalues

The question. How can a link network determine a unique probability ranking? Plain transitions may fail: I_2 has many stationary distributions, and the swap matrix has a unique stationary distribution but alternates e_1, e_2.

Definition (Notes Chapter 11: link transitions)**.** For n pages, a directed edge $i \to j$ means page i links to page j. The adjacency matrix is $C_{ij} = 1$ when that edge exists and 0 otherwise. A directed walk of length k follows k consecutive edges, with repetitions allowed. Let $d_i = \sum_j C_{ij}$ be the number of outgoing links. When all $d_i > 0$, put $D = \operatorname{diag}(d_1, \ldots, d_n)$, $A = D^{-1}C$, and $P = A^{\mathsf{T}} = C^{\mathsf{T}}D^{-1}$.

Proposition (Notes Chapter 11: walks and transition conventions)
$(C^k)_{ij}$ counts length-k walks from i to j for $k \geq 1$. The matrix A is row-stochastic and P is column-stochastic, with $P_{ij} = 1/d_j$ if $j \to i$ and 0 otherwise. Thus P describes choosing an outgoing link uniformly from the current page.

Where the proof comes from

Want. Relate the network description to the course column-vector convention.

Know. Matrix multiplication sums over the intermediate index.

Key step. Partition walks by their penultimate vertex and normalize each outgoing row.

Proof. 1. For $k = 1$, C_{ij} counts the possible single edge. If the formula holds for k, a length-$(k+1)$ walk to j consists uniquely of a length-k walk to some ℓ followed by edge $\ell \to j$. Its count is $\sum_\ell (C^k)_{i\ell} C_{\ell j} = (C^{k+1})_{ij}$.

2. Row i of $D^{-1}C$ has d_i entries equal to $1/d_i$, hence nonnegative entries summing to 1. Transposing makes these the columns of P, proving the stated entries and column sums.

□

Definition (Notes Chapter 11: damping and dangling pages)**.** A page with no outgoing links is dangling. For a chosen probability vector v, define column j by $P_{ij} = C_{ji}/d_j$ when $d_j > 0$, and set column j equal to v when $d_j = 0$. Thus P is column-stochastic even with dangling pages. Fix $0 < \alpha < 1$, the damping parameter. PageRank seeks r satisfying $r = \alpha P r + (1 - \alpha)v$: with probability α follow a link, and with probability $1 - \alpha$ choose a page from the restart distribution v.

Definition (Notes Chapter 11: the 1-norm)**.** For $x \in \mathbb{C}^n$, define $\|x\|_1 = \sum_j |x_j|$.

Proposition (Notes Chapter 11: 1-norm properties)
$\|x\|_1 = 0$ iff $x = 0$, $\|ax\|_1 = |a|\|x\|_1$, and $\|x + y\|_1 \leq \|x\|_1 + \|y\|_1$.

Proof. 1. A sum of nonnegative moduli is zero exactly when each coordinate is zero.
2. $\|ax\|_1 = \sum_j |ax_j| = |a| \sum_j |x_j| = |a|\|x\|_1$.
3. Summing $|x_j + y_j| \le |x_j| + |y_j|$ over j gives the triangle inequality.

□

Proposition (Notes Proposition 57)

For a column-stochastic matrix P and any $x \in \mathbb{C}^n$, $\|Px\|_1 \le \|x\|_1$. Every complex eigenvalue λ of P satisfies $|\lambda| \le 1$.

Where the proof comes from

Want. Bound eigenvalues without computing them.

Know. Entries of P are nonnegative and each column sums to 1.

Key step. Absolute values prevent direct cancellation in each output coordinate. First bound $|\sum_j P_{ij}x_j| \le \sum_j P_{ij}|x_j|$, then sum over i. Each $|x_j|$ is multiplied by column sum $\sum_i P_{ij} = 1$, giving $\|Px\|_1 \le \|x\|_1$.

Proof. 1. $\|Px\|_1 = \sum_i |\sum_j P_{ij}x_j| \le \sum_{i,j} P_{ij}|x_j| = \sum_j (\sum_i P_{ij})|x_j| = \|x\|_1$.
2. If $Px = \lambda x$ and $x \ne 0$, then $|\lambda|\|x\|_1 = \|Px\|_1 \le \|x\|_1$. Divide by the positive quantity $\|x\|_1$ to obtain the bound.

□

Series tools (cited elementary analysis). For $0 < \alpha < 1$, $\sum_{k=0}^{\infty} \alpha^k = 1/(1-\alpha)$ and $\alpha^k \to 0$. A series dominated termwise in absolute value by a convergent nonnegative series converges; finite linear combinations commute with these limits. These facts justify the coordinate series below.

Proposition (Notes Proposition 58: existence and uniqueness of PageRank)

For a column-stochastic P, probability vector v, and $0 < \alpha < 1$, there is a unique $r \in \mathbb{R}^n$ with $r = \alpha Pr + (1-\alpha)v$. It is a probability vector, with $r = (1-\alpha)\sum_{k=0}^{\infty} \alpha^k P^k v$.

Where the proof comes from

Want. Prove uniqueness, existence, nonnegativity, and total probability 1.

Know. Proposition 57 bounds eigenvalues; Week 4 turns injectivity into invertibility in finite dimension; stochastic maps preserve probability vectors.

Key step. The eigenvalue bound excludes $1/\alpha$, so $I - \alpha P$ is invertible. This gives a unique solution but says nothing about its signs. Substituting $r = \alpha Pr + (1-\alpha)v$ repeatedly suggests $r = (1-\alpha)\sum_{k\ge 0} \alpha^k P^k v$. Every $P^k v$ is a probability vector, and these nonnegative geometric weights sum to 1, explaining why this expression proves nonnegativity and normalization.

Proof. 1. If $(I - \alpha P)x = 0$ and $x \ne 0$, then $Px = \alpha^{-1}x$ would give eigenvalue $\alpha^{-1} > 1$, contradicting Proposition 57. Thus $I - \alpha P$ has zero kernel and is invertible.
2. Therefore $(I - \alpha P)r = (1-\alpha)v$ has exactly one real solution. To identify its signs, set $s = (1-\alpha)\sum_{k\ge 0} \alpha^k P^k v$.
3. Each $P^k v$ is a probability vector, so each coordinate is between 0 and 1. Thus each coordinate series converges by comparison with $(1-\alpha)\sum \alpha^k$, and every coordinate of s is nonnegative.
4. Let $\mathbf{1} = (1, \dots, 1)^\mathsf{T}$. Then $\mathbf{1}^\mathsf{T} s = (1-\alpha)\sum_{k\ge 0} \alpha^k \mathbf{1}^\mathsf{T} P^k v = (1-\alpha)\sum_{k\ge 0} \alpha^k = 1$.
5. Shifting the absolutely convergent coordinate series gives $\alpha Ps + (1-\alpha)v = (1-\alpha)\sum_{k\ge 1} \alpha^k P^k v + (1-\alpha)v = s$. By uniqueness $s = r$, proving the assertions.

□

Proposition (Notes Chapter 11: iterative convergence)

From any initial probability vector r_0, define $r_{k+1} = \alpha Pr_k + (1-\alpha)v$. Every iterate is a probability vector and $\|r_k - r\|_1 \le \alpha^k \|r_0 - r\|_1$, hence $r_k \to r$. The map $F(x) = \alpha Px + (1-\alpha)v$ is a contraction: $\|F(x) - F(y)\|_1 \le \alpha\|x - y\|_1$.

Proof. 1. If r_k is a probability vector, so are Pr_k and v; the weighted sum with nonnegative weights $\alpha, 1-\alpha$ has nonnegative entries and sum 1. Induction proves the first claim.

2. $r_{k+1} - r = \alpha P(r_k - r)$, so Proposition 57 gives $\|r_{k+1} - r\|_1 \le \alpha \|r_k - r\|_1$. Induction yields the bound, and the cited geometric limit gives convergence.
3. For arbitrary x, y, $F(x) - F(y) = \alpha P(x - y)$, giving the contraction inequality. A contraction here means exactly such an inequality with a constant less than 1; no general fixed-point theorem is needed for this proof.

□

Definition (Notes Chapter 11: Google matrix)**.** Let $G = \alpha P + (1-\alpha) v \mathbf{1}^\mathsf{T}$ and $H = \{x \in \mathbb{R}^n : \mathbf{1}^\mathsf{T} x = 0\}$.

Proposition (Notes Chapter 11: invariant decomposition for PageRank)

G is column-stochastic and $Gx = \alpha P x + (1-\alpha) v$ for probability vectors x, so $Gr = r$. The space H is invariant under both P and G, with $G|_H = \alpha P|_H$. Moreover $\mathbb{R}^n = \operatorname{span}(r) \oplus H$, and G is the identity on $\operatorname{span}(r)$.

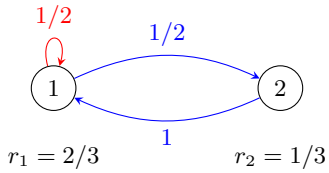
Proof. 1. Each column of $v\mathbf{1}^\mathsf{T}$ equals v, so each column of G is a probability vector by its weights $\alpha, 1-\alpha$. For probability x, $\mathbf{1}^\mathsf{T} x = 1$, giving $Gx = \alpha P x + (1-\alpha) v$ and hence $Gr = r$.

2. H is the kernel of the linear functional $x \mapsto \mathbf{1}^\mathsf{T} x$, hence a subspace. For $x \in H$, $\mathbf{1}^\mathsf{T} P x = \mathbf{1}^\mathsf{T} x = 0$, so $Px \in H$.
3. For $x \in H$, $Gx = \alpha P x + (1-\alpha) v \mathbf{1}^\mathsf{T} x = \alpha P x \in H$, proving both the invariance and restriction formula.
4. Every x equals $(\mathbf{1}^\mathsf{T} x) r + [x - (\mathbf{1}^\mathsf{T} x) r]$; the bracketed vector has coordinate sum 0 and lies in H.
5. If $cr \in H$, then $0 = \mathbf{1}^\mathsf{T}(cr) = c$, so the intersection is zero. This proves directness. Since $Gr = r$, linearity gives $G(cr) = cr$.

□

Compute PageRank in a two-state example

1. Take the graph $1 \to 2$, $2 \to 1$. Its adjacency and transition matrices coincide: $C = P = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, and $C^2 = I$ counts one length 2 walk from each state to itself. Choose $v = e_1$ and $\alpha = 1/2$. The equations are $r_1 = \frac{1}{2} r_2 + \frac{1}{2}$, $r_2 = \frac{1}{2} r_1$.
2. Substitution gives $r = (2/3, 1/3)^\mathsf{T}$. The Google matrix is $G = \left(\begin{smallmatrix} 1/2 & 1 \\ 1/2 & 0 \end{smallmatrix}\right)$, and multiplication verifies $Gr = r$.
3. From $r_0 = e_1$, the first iterates are $(1/2, 1/2)^\mathsf{T}$ and $(3/4, 1/4)^\mathsf{T}$; their 1-norm errors are $1/3$ and $1/6$, exactly half at each step in this example.



These are the transitions of G, after the restart rule is included. From state 1, half the probability stays and half moves to state 2; state 2 always moves to state 1. At this distribution, the probability moving from 1 to 2 equals that moving from 2 to 1: $(2/3)(1/2) = (1/3)1$.

Historical context from the notes. Page and Brin developed PageRank as a link-based component of search ranking; query relevance was a separate component. The often-used damping value 0.85 and uniform restart vector describe an early convention, not a claim about present search-engine parameters. A nonuniform v gives a personalized restart distribution.

Shortcuts

- Solve $(I - \alpha P) r = (1-\alpha) v$ directly for small examples; invertibility is already guaranteed.
- For iteration, the error decreases by at least a factor α in the 1-norm, providing a bound without computing eigenvectors.
- Check a candidate by its fixed-point equation, nonnegative entries, and coordinate sum 1.

Traps

- A stochastic matrix need not have a unique stationary distribution: $P = I_2$ fixes every probability vector.
- An eigenvector for 1 need not be a probability vector: $I_2(1,-1) = (1,-1)$, but its coordinates are not nonnegative.
- Unique stationary distribution does not imply convergence: the swap matrix fixes only $(1/2, 1/2)$ among probability vectors but alternates e_1, e_2.
- A PageRank vector need not be strictly positive if v has zero entries: $P = I_2, v = e_1$ gives $r = e_1$.
- Do not divide by zero outgoing degree: a dangling page requires replacing its transition column before the stochastic arguments apply.

§6 Problem set

Operator convention. In HW6 P1, P3 and P6 the source omits some explicit linearity assumptions. The intended maps P, S, T are linear operators. Without that assumption, P1 is false: on $\mathbb{R}$, $P(x) = |x|$ satisfies $P^2 = P$ but its range $[0, \infty)$ is not a subspace. The original idempotent equation is retained below with the needed operator hypothesis stated.

HW6 P1. Let the linear operator $P : V \to V$ satisfy $P^2 = P$. (a) Prove each eigenvalue is in $\{0, 1\}$. (b) Prove $V = \ker P \oplus \operatorname{range} P$.
Hint: Apply $P^2 = P$ to an eigenvector; decompose $v = (v - Pv) + Pv$.

HW6 P2. For differentiation $D : \mathcal{P}_n(F) \to \mathcal{P}_n(F)$, $F = \mathbb{R}$ or $\mathbb{C}$, (a) find all eigenvalues and eigenspaces. (b) Explain the difference on smooth functions on $\mathbb{R}$.
Hint: Compare degrees for polynomials; use exponential functions for smooth eigenfunctions.

HW6 P3. Let $S, T \in L(V)$ with S invertible and put $R = S^{-1}TS$. Prove $E(\lambda, R) = S^{-1}E(\lambda, T)$ for every scalar λ, hence equal sets of eigenvalues.
Hint: Multiply the equation $Rv = \lambda v$ by S.

HW6 P4. For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, use only the two-by-two determinant formula to prove $\det(tI - A) = t^2 - (a+d)t + (ad - bc)$. Prove λ is an eigenvalue iff it is a root of this polynomial.
Hint: Expand the two-by-two determinant and use failure of invertibility.

HW6 P5. Find eigenvalues, eigenspaces, and whether an eigenvector basis exists for $A_1 = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ first over $\mathbb{R}$ then $\mathbb{C}$; $A_2 = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ over $\mathbb{R}$; and $A_3 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ over $\mathbb{R}$.
Hint: Use the two-by-two characteristic polynomial, then solve each kernel.

HW6 P6 (optional). For $T \in L(V)$ and $v \neq 0$, define the cyclic subspace $C(v, T) = \operatorname{span}(v, Tv, T^2v, \ldots)$, meaning finite combinations of the listed iterates. (a) Prove it is invariant. (b) If V is finite-dimensional, show $C(v, T) = \operatorname{span}(v, \ldots, T^{m-1}v)$ for some $m \leq \dim V$.
Hint: Take the first dependent initial list, then show the preceding span is preserved.

HW6 P7 (optional). For $P = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1/2 & 1/2 \end{pmatrix}$, find a probability vector p with $Pp = p$ and explain its Markov-chain meaning.
Hint: Solve the first and third stationary equations, then normalize the sum.

Exam practice (midterm2-review.pdf, Problem 7). For $T \in L(V)$, prove $\ker T$ and $\operatorname{range} T$ are invariant under T. Further prove their annihilators are invariant under the dual map T'. Recall from Week 5 that $V' = L(V, F)$, $U^0 = \{\varphi \in V' : \varphi(u) = 0 \text{ for every } u \in U\}$ and $(T'\varphi)(v) = \varphi(Tv)$.
Hint: For any invariant U, evaluate $(T'\varphi)(u) = \varphi(Tu)$ with $\varphi \in U^0$.